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Flow over and forced convection heat transfer in Newtonian fluids from a semi-circular cylinder

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ABSTRACT

Momentum and heat transfer characteristics of a semi-circular cylinder immersed in unconfined flowing Newtonian fluids have been investigated numerically. The governing equations, namely, continuity, Navier–Stokes and energy, have been solved in the steady flow regime over wide ranges of the Reynolds number ($0.01 \leqslant Re \leqslant 39.5$) and Prandtl number ($Pr \leqslant 100$). Prior to the investigation of drag and heat transfer phenomena, the critical values of the Reynolds number for wake formation ($0.55 < Re^c < 0.6$) and for the onset of vortex shedding ($39.5 < Re_c < 40$) have been identified. The corresponding values of the lift coefficient, drag coefficient, and Strouhal number are also presented. After establishing the limit of the steady flow regime, the influence of the Reynolds number ($0.01 \leqslant Re \leqslant 39.5$) and Prandtl number ($Pr = 0.72, 1, 10, 50$ and 100) on the global flow and heat transfer characteristics have been elucidated. Detailed kinematics of the flow is investigated in terms of the streamline and vorticity profiles and the variation of pressure coefficient in the vicinity of the cylinder. The functional dependence of the individual and total drag coefficients on the Reynolds number is explored. The Nusselt number shows an additional dependence on the Prandtl number. In addition, the isotherm profiles, local Nusselt number (Nu_L) and average Nusselt number (Nu) are also presented to analyze the heat transfer characteristic of a semi-circular cylinder in Newtonian media.

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1. Introduction

Over the years, considerable research efforts have been directed at furthering our understanding of the flow of fluids past cylinders of various cross-sections, especially circular, elliptic and square/rectangular cylinders. The interest in such model studies stems from both theoretical and pragmatic considerations. From a theoretical point, bluff body flows constitute an important class of problems within the domain of fluid mechanics which have proved to be of value to gain useful insights into the nature of underlying phenomena such as wake characteristics, vortex shedding, etc. which often serve as a launching pad to investigate other body shapes as well as interactions between multiple bluff bodies as encountered in numerous applications. From a practical standpoint, the flow past a single cylinder denotes an idealization of several industrially important processes. Typical examples include the flow in tubular and pin-type heat exchangers, processing of fibrous suspensions as encountered in paper making and in fiber-reinforced polymer composites, for instance. Further practical applications are found in the use of screens to dewater coal-water slurries, filtration of sewage sludge and polymer melts, removal of oversized particles from coating suspensions. Consequently, over the years, significant research efforts have been expended in studying momentum and heat transfer characteristics from cylinders of various cross-sections including circular, elliptic, and rectangular in particular. Indeed, the available voluminous literature has been reviewed succinctly in excellent books [1–3] and survey articles [4,5]. An inspection of these resources clearly suggests that the circular cylinder has received maximum attention followed by non-circular (elliptic, square, rectangular) shapes. But nonetheless an adequate body of knowledge is now available encompassing various aspects of this general class of flow over wide ranges of conditions.

In contrast, the flow over a semi-circular cylinder has received very little attention. The flow over a semi-circular cylinder is relevant in the development and design of novel heat exchangers (improved heat transfer characteristics without a severe pressure drop penalty) and in the thermal processing of foodstuffs (carrot/potato chips, for instance) [6]. Further applications are found in the cooling of electronic components and chips of various shapes, underwater vessels like submarines with a flat bottom, the use of such objects to form weld-lines in polymer shaping operations, etc. Finally, apart from its aforementioned pragmatic significance, this flow deserves attention in its own right as one of the fundamental problems in fluid dynamics.

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Nomenclature

C_D	total drag coefficient (dimensionless)
$\overline{C}_D$	time-averaged drag coefficient (dimensionless)
C_{DP}	pressure drag coefficient (dimensionless)
C_{DF}	friction drag coefficient (dimensionless)
C_L	total lift coefficient (dimensionless)
c_p	specific heat of the fluid (J/kg K)
C_p	pressure coefficient (dimensionless)
D	diameter of cylinder (m)
D_∞	diameter of outer domain (m)
e	aspect ratio of elliptic cylinder (ratio of minor to major axis)
f	frequency of vortex shedding (s^{-1})
F_D	total drag force per unit length of cylinder (N/m)
F_{DF}	friction drag force per unit length of cylinder (N/m)
F_{DP}	pressure drag force per unit length of cylinder (N/m)
F_L	total lift force per unit length of cylinder (N/m)
h	local convective heat transfer coefficient (W/m^2 K)
k	thermal conductivity of fluid (W/m K)
L	distance from the rear stagnation point (the center of the cylinder) to the point of reattachment for the near closed streamline (m)
L_r	recirculation length (dimensionless)
n_s	unit vector normal to the surface of the cylinder
n_x	x-component of the direction vector normal to the surface of the cylinder
Nu	Nusselt number (dimensionless)
Nu_L	local Nusselt number (dimensionless)
Nu_{avg}	average Nusselt number (dimensionless)
p	pressure (Pa)
p_s	local surface pressure (Pa)
p_∞	free stream pressure (Pa)
Pr	Prandtl number (dimensionless)
Re	Reynolds number (dimensionless)
Re^c	critical Reynolds number for onset of flow separation (dimensionless)
Re_c	critical Reynolds number for onset of vortex shedding (dimensionless)
S	surface area (m^2)
T	fluid temperature (K)
T_∞	fluid temperature at inlet (K)
T_w	wall temperature at the surface of the semi-circular cylinder (K)
U_∞	free stream velocity at the inlet (m/s)
U_x	x-component of the velocity (m/s)
U_y	y-component of the velocity (m/s)

Greek symbols

ε	rate of strain tensor (s^{-1})
δ	grid spacing at the vicinity of the cylinder
δ_T	thermal boundary layer thickness (m)
φ	U_x, U_y and T
μ	viscosity of the fluid (Pa s)
ρ	density of the fluid (kg/m^3)
τ	extra stress (Pa)

Subscripts

i, j, x, y	Cartesian coordinates

Superscript

$'$	dimensional variable

It is readily conceded that the flow over a bluff body (and hence the rate of interphase heat/mass transport) is influenced by a large number of variables including the shape and orientation of the bluff body, nature of oncoming flow (shear or uniform), type of fluid (compressible or incompressible), angle of incidence, confined or unconfined, etc. Thus, for instance, for an unconfined circular cylinder, the flow is steady and does not separate from the surface of the cylinder up to about $Re \sim 4$–5. Therefore, the flow exhibits fore and aft as well as mid-plane symmetry in this flow regime up to about $Re \leqslant \sim$4–5. As the value of the Reynolds number is progressively increased, the flow separates due to the occurrence of an adverse pressure gradient at about $Re \sim 5$–6. The size of the symmetrical wake in the form of two standing vortices increases with Reynolds number. Though the flow does not exhibit the fore and aft symmetry any longer, but it is still symmetric about the mid-plane. This flow regime continues up to about $Re \sim 47$ beyond which the flow becomes asymmetric and time dependent. The change in the flow behaviour also modifies the nature of the functional dependence of engineering parameters (drag, lift, Nusselt number, etc.) on the Reynolds and Prandtl numbers. For a circular cylinder, the separation point moves towards the front stagnation point as the value of the Reynolds number is gradually increased. Naturally, the transition from one flow regime to another is strongly influenced not only by the shape of the cylinder, but also by its orientation (with respect to the oncoming flow) and whether the flow is confined or unconfined. A circular cylinder represents the simplest bluff body shape, which is free from geometric singularities. All other shapes have an additional complication due to either the presence of sharp corners, or lack of symmetry and hence due to its orientation with respect to the flow direction. One would expect the occurrence of similar different flow regimes for the flow over a semi-circular cylinder also under appropriate condition. Despite such an overwhelming fundamental and practical significance, a brief examination of the available literature suggests that very little is known about the flow of fluids past a semi-circular cylinder. This work is concerned with the low Reynolds number flow over a semi-circular cylinder oriented normal to the direction of flow with its curved surface facing upstream. The prevailing Reynolds numbers in food processing and polymer processing applications are likely to be rather small. This is so due to the relatively high viscosity of most polymeric liquids, and/or by design to minimize the damage to product quality in food processing applications. However, at the outset, it is instructive and useful to briefly review the limited available literature on this flow configuration, which, in turn, facilitates the presentation and discussion of the new results obtained in this study for a semi-circular cylinder.

2. Previous work

As noted in the preceding section, excellent reviews are available for the flow of Newtonian fluids over cylinders of circular, square, and elliptic cross-sections [1–5]. Most of the recent studies on the fluid mechanical and heat transfer aspects in the flow past a square cylinder in the 2-D laminar flow have been summarized in some of our recent studies [7–9]. Suffice it to say here that an adequate body of knowledge is also available on the flow past a square cylinder in the laminar flow regime. The effect of Prandtl number on heat transfer has also been explored recently and the results encompassing a 100-fold variation in the value of Prandtl number

conform to the expected scaling of the Nusselt number $Nu \sim Pr^{1/3}$ [7]. During the last 10–15 years, there has been a growing interest in studying the flow of non-Newtonian, especially power-law type fluids, over a circular [10–12], elliptic [13–15] and square [7–9] cylinders. Most of these studies are restricted to the so-called steady flow regime, with the notable exception of limited time dependent simulations reported recently for a circular cylinder [16,17] and for a square cylinder [18–21]. Broadly speaking, all else being equal, shear-thinning behaviour enhances the hydrodynamic drag and the rate of heat transfer with reference to that in Newtonian fluids. Shear-thickening has the opposite influence. However, this effect is particularly striking at low Reynolds numbers. This is simply so due to the fact that as the value of Reynolds number is gradually incremented, the flow is dominated by the inertial forces and therefore the type of viscous behaviour of a fluid is of little relevance under these conditions. Overall, the literature on the effect of non-Newtonian fluid behaviour in bluff body flows is neither as extensive nor as conclusive as that for Newtonian fluids.

In contrast, the corresponding body of knowledge for a semi-circular cylinder is much less extensive, even for Newtonian fluids. The available literature comprises numerical and experimental studies for different orientations of a semi-circular cylinder with respect to the direction of the mean flow. Kiya and Arie [22] obtained a numerical solution of the Navier–Stokes equations for the 2-D viscous shear flow over a semi-circular and semi-elliptical projection attached to a plane wall for the Reynolds number values ranging from 0.1 to 100. They reported extensive results on geometrical shape of the front and rear vortices, drag coefficients and pressure and shear distribution as a function of the Reynolds number. The dependence of the flow behaviour of a semi-circular obstacle in the ideal 2D-critical free surface flow has also been explored by Forbes and co-workers [23,24]. They found that the downstream velocity increases with the radius of semi-circular obstacle. Boisaubert et al. [25] reported an experimental study of 3-D confined flow over a semi-circular cylinder for Reynolds number ranging from 60 to 600 for the flat and curved surfaces facing the flow using a solid tracer visualization technique. They also reported the critical values of the Reynolds numbers for the onset of vortex shedding as $Re = 140$ and $Re = 190$ for the two configurations respectively. Subsequently, they introduced a splitter plate behind the rounded forebody configuration to study its effect on the flow behaviour, and suggested the suitability of this arrangement as a flow-controlling device [26,27]. The variation of the stagnation points and the behaviour of vortices for a semi-circular cylinder exposed to an uniform shear flow have also been studied by Kotake and Suwa [28] using the hydrogen bubble method (visualization technique) and found that the vortex was formed only on the side of the body where the fluid velocity was lower. Iguchi and Terauchi [29] reported that the Karman vortex street is significantly influenced by the attack angle measured from the flow direction. Koide et al. [30] carried out an experimental investigation of the influence of circular, semi-circular and triangular cross-section configuration of a cylindrical body on the lock-in (when the frequency of vortex shedding coincides with that of free oscillating cylinder) phenomenon of Karman vortex shedding using mechanical oscillator for producing cross flow oscillations. They found that the separation point could move freely for a circular cylinder, its movement is restricted to the upstream curved surface for a semi-circular cylinder and fixed at the rear vertices for a triangular cylinder (apex facing upstream). The value of Reynolds number $Re > 3500$ in this study is way beyond the limits of the currently available numerical simulations of any of the shapes (with the exception of a circular cylinder) considered herein. Sophy et al. [31] found the flow to be unsteady at Reynolds number of 65 for the flow past a semi-circular cylinder (curved surface facing the flow) and the corresponding Strouhal number was found to be 0.166, which is 7% larger than the case of a circular cylinder at the corresponding transition. On the other hand, Nada and Mowad [32] reported experimental results on free convection heat transfer from a semi-circular cylinder at different orientations. Subsequently, they reported experimental and numerical results for forced convection and fluid flow around a semi-circular cylinder for the range $Re = 2.2 \times 10^3$–4.5×10^4 and orientation angle of 30, 60, and 90 from flow direction [33]. They also presented correlations for the average Nusselt number in terms of Reynolds number and angle of attack. Such high values of the Reynolds number are also beyond the limits of the currently available numerical predictions. Hocking and Vanden-Broeck [34] have explored the effect of gravity on the wake region of the flow past a semi-circular cylinder, when the flow is parallel to the flat surface. They found narrow and long wake regions for the upward facing curved surface configuration and wide and short wake region for the downward facing curved surface configuration. Finally, some results on the size of the recirculation zone and formation of vortices are also available on a hollow semi-circular cylinder with the curved surface facing the flow [35,36]. Evidently, although this flow is much more complex than the one studied here, it is included here for the sake of completeness.

From the foregoing description, it is thus safe to conclude that very little is known about the flow past a semi-circular cylinder and no systematic effort seems to have been devoted to develop a rationale understanding of the underlying physical phenomena related to momentum and heat transfer characteristics. This work aims to alleviate this situation for the case when the curved surface is facing upstream. In particular, consideration is given to the delineation of the critical values of the Reynolds number denoting the onset of flow separation and of the laminar vortex shedding regimes. In addition, extensive numerical results are obtained for the detailed kinematics and gross flow and heat transfer parameters in the steady flow regime over wide ranges of conditions as: Reynolds number, 0.01 to ~40 and for Prandtl numbers ranging from 0.72 to 100.

3. Problem statement and mathematical formulation

Consider the two-dimensional, laminar flow of an incompressible Newtonian fluid with a uniform inlet velocity U_∞ and temperature, T_∞ across an infinitely long (in z-direction) semi-circular cylinder (Fig. 1(a)). Strictly speaking, it is not feasible to simulate the unconfined flow condition numerically, so it is customary to confine the flow using artificial boundaries having negligible effect on the flow characteristics. To convert the physical problem into a computational equivalent, a semi-circular cylinder of diameter D is placed concentrically in a circular domain of diameter D_∞ as shown in Fig. 1(b). The radius of the enveloping circular domain is chosen to be sufficiently large in order to minimize the boundary effects. The surface of the solid cylinder is maintained at a constant wall temperature, T_w. The thermo-physical properties (viscosity, μ, density, ρ, heat capacity, c_p and thermal conductivity, k) of the streaming liquid are assumed to be independent of the temperature and the viscous dissipation effects in the energy equation are neglected in this study. While these two assumptions lead to the decoupling of the momentum and energy equations, the first assumption also restricts the applicability of the present results to the situations wherein the temperature difference $(T_w - T_\infty)$ is small. Under these assumptions, the flow and heat transfer phenomena are governed by the continuity, Navier–Stokes and thermal energy equations written in their dimensionless forms, as follows.

Continuity equation:

$$\frac{\partial U_x}{\partial x} + \frac{\partial U_y}{\partial y} = 0 \tag{1}$$

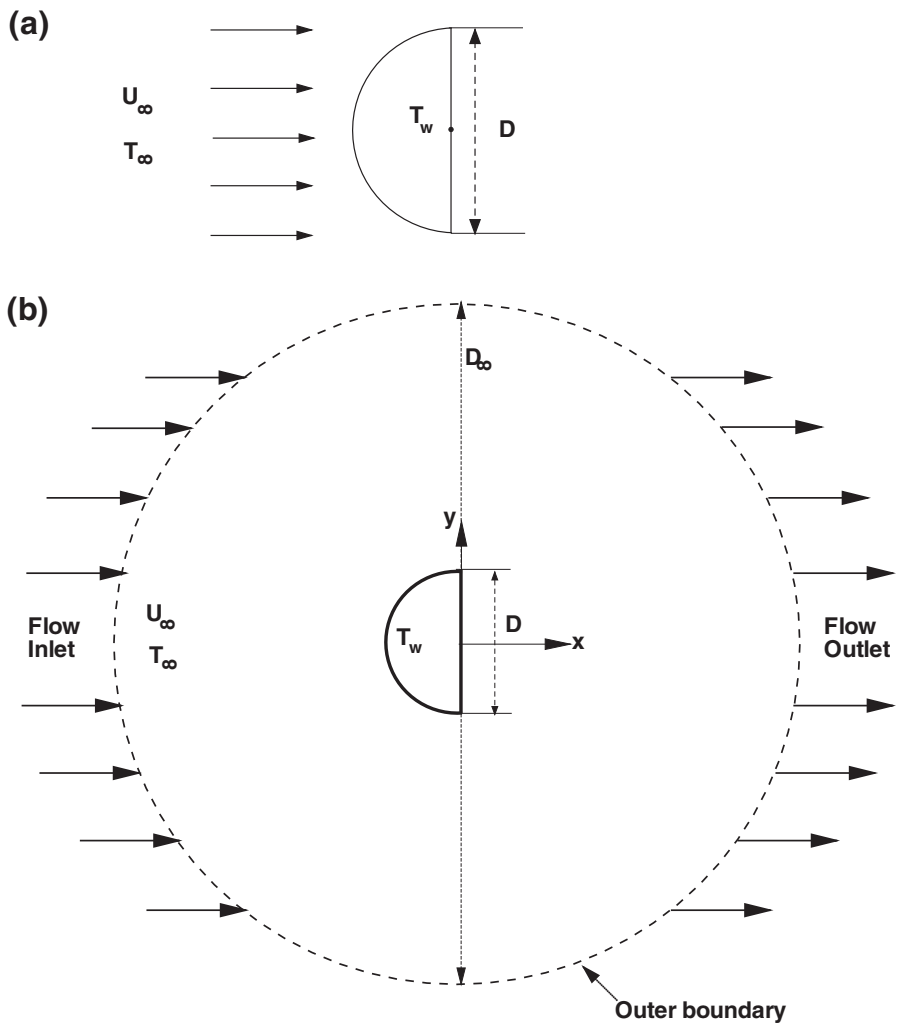

Fig. 1. Schematics of the unconfined flow around a semi-circular cylinder: (a) physical model; (b) computational domain.

Navier–Stokes equation:

x-component:

$$\frac{DU_x}{Dt} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 U_x \tag{2}$$

y-component:

$$\frac{DU_y}{Dt} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\nabla^2 U_y \tag{3}$$

Energy equation:

$$\frac{DT}{Dt} = \frac{1}{Re \times Pr}\nabla^2 T \tag{4}$$

The variables appearing in the aforementioned equations are rendered dimensionless using $D, U_\infty, D/U_\infty$, and ρU_∞^2 as scaling variables for length, velocity, time and pressure respectively. The temperature is made dimensionless as $\frac{T'-T_\infty}{T_w-T_\infty}$. The two dimensionless groups appearing in Eq. (4) are defined as follows:

$$Re = \frac{\rho D U_\infty}{\mu} \tag{5}$$

$$Pr = \frac{c_p \mu}{k} \tag{6}$$

The physically realistic boundary conditions in their non-dimensional forms for this flow may be written as follows:

At the inlet boundary: An uniform inlet velocity in *x*-direction and constant temperature, T_∞ are prescribed, *i.e.*,

$$U_x = 1; \quad U_y = 0; \quad T = 0 \tag{7}$$

At the surface of the cylinder: The usual no-slip boundary condition for flow and that of constant temperature, T_w, for heat transfer are implemented, *i.e.*,

$$U_x = 0; \quad U_y = 0; \quad T = 1 \tag{8}$$

At the exit boundary: The outflow boundary condition is used as:

$$\frac{\partial \varphi}{\partial x} = 0 \tag{9}$$

where $\varphi = U_x, U_y$ and T.

This is the default outflow boundary condition provided in FLUENT (6.3.26) and it is appropriate under the present circumstances

Table 1
Variation of drag coefficient with domain size.

Domain size (D_∞/D)	$Re = 0.01$			$Re = 5$		
	C_{DP}	C_{DF}	C_D	C_{DP}	C_{DF}	C_D
1200	265.51	140.88	406.39	2.335	1.482	3.817
1400	261.43	138.71	400.14	2.334	1.482	3.816
1600	258.46	136.96	395.42	–	–	–
1800	256.05	135.68	391.73	–	–	–

where the exit flow is close to a fully developed condition. It is equivalent to the homogeneous Neumann boundary condition and assumes zero diffusion flux for all flow variables in the axial direction, albeit the gradients in the lateral direction can still exist.

The solution to Eqs. (1)–(4) maps the flow domain in terms of the primitive variables, *i.e.*, U_x and U_y, pressure, p and temperature, T. These, in turn, can be post processed to deduce the quantities of interest like drag and lift coefficients, stream function, Nusselt number, etc. as functions of the Reynolds and Prandtl numbers. It is therefore appropriate to briefly introduce the evaluation of these quantities here.

3.1. Total drag and lift coefficient

The total drag coefficient is the sum of the friction and pressure drag coefficients. These are evaluated as follows:

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 D} = C_{DP} + C_{DF} \tag{10}$$

$$C_{DP} = \frac{F_{DP}}{\frac{1}{2}\rho U_\infty^2 D} = \int_S C_P n_x dS \tag{11}$$

$$C_{DF} = \frac{F_{DF}}{\frac{1}{2}\rho U_\infty^2 D} = \frac{2}{\mathrm{Re}} \int_S (\tau \cdot n_s) dS \tag{12}$$

where n_s is the unit normal vector on the surface of the semi-circular cylinder and n_x is its x-component.

In the time-periodic flow regime, there is also a net force acting on the cylinder in the lateral direction and this is expressed in terms of a lift coefficient C_L, defined as follows:

$$C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 D} \tag{13}$$

3.2. Strouhal number

Strouhal number is the dimensionless frequency of vortex shedding and it is defined here as

$$St = \frac{fD}{U_\infty} \tag{14}$$

Table 2
Dependence of drag coefficients on grid size.

Grid	δ/D	Points on cylinder surface	Total cells	$Re = 0.01$			$Re = 5$		
				C_{DP}	C_{DF}	C_D	C_{DP}	C_{DF}	C_D
*G*1	0.02	130	18566	258.46	136.96	395.42	2.335	1.482	3.817
*G*2	0.01	258	47051	264.38	131.97	396.35	2.376	1.450	3.826
*G*3	0.005	514	148555	268.94	128.19	397.13	2.404	1.429	3.832
*G*4	0.002	1284	221056	272.29	125.42	397.71	2.426	1.413	3.839

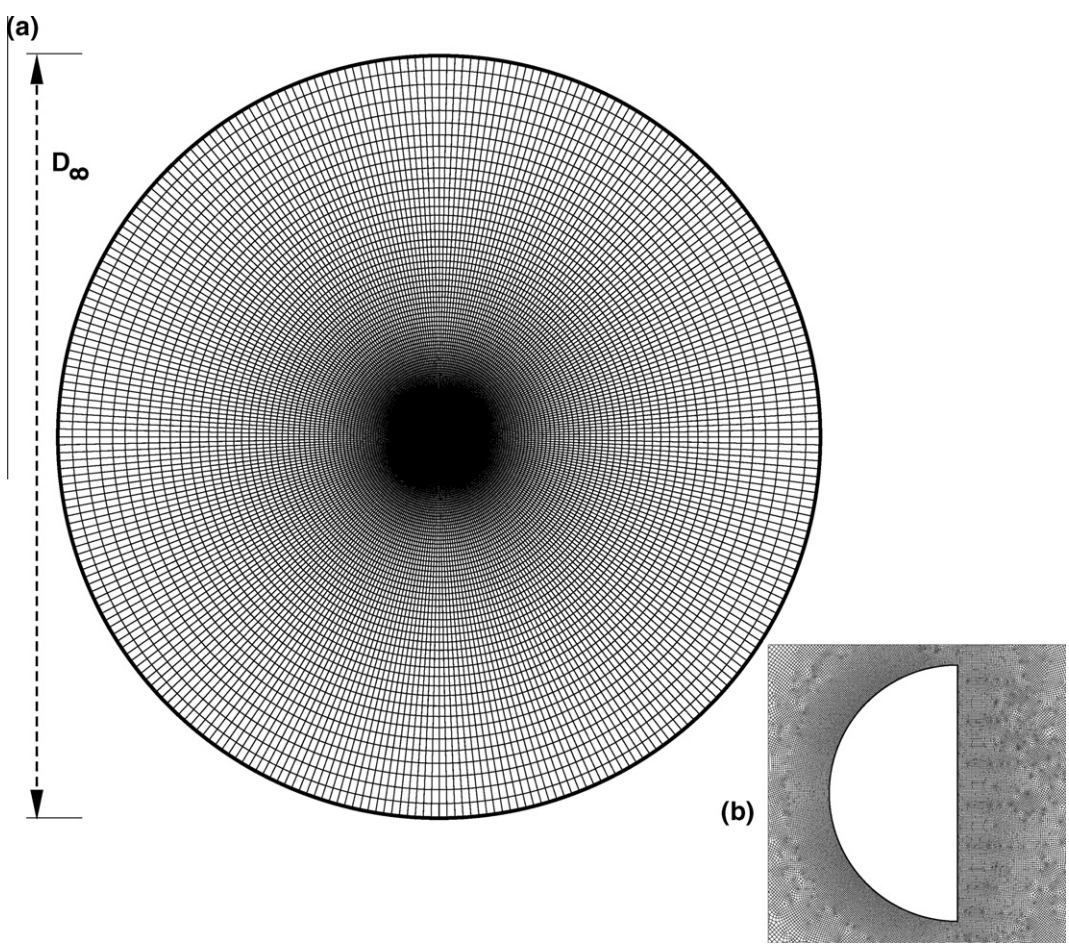

Fig. 2. Schematic representation of non-uniform 2-D computational grid structure: (a) full grid; (b) grid near the semi-circular cylinder.

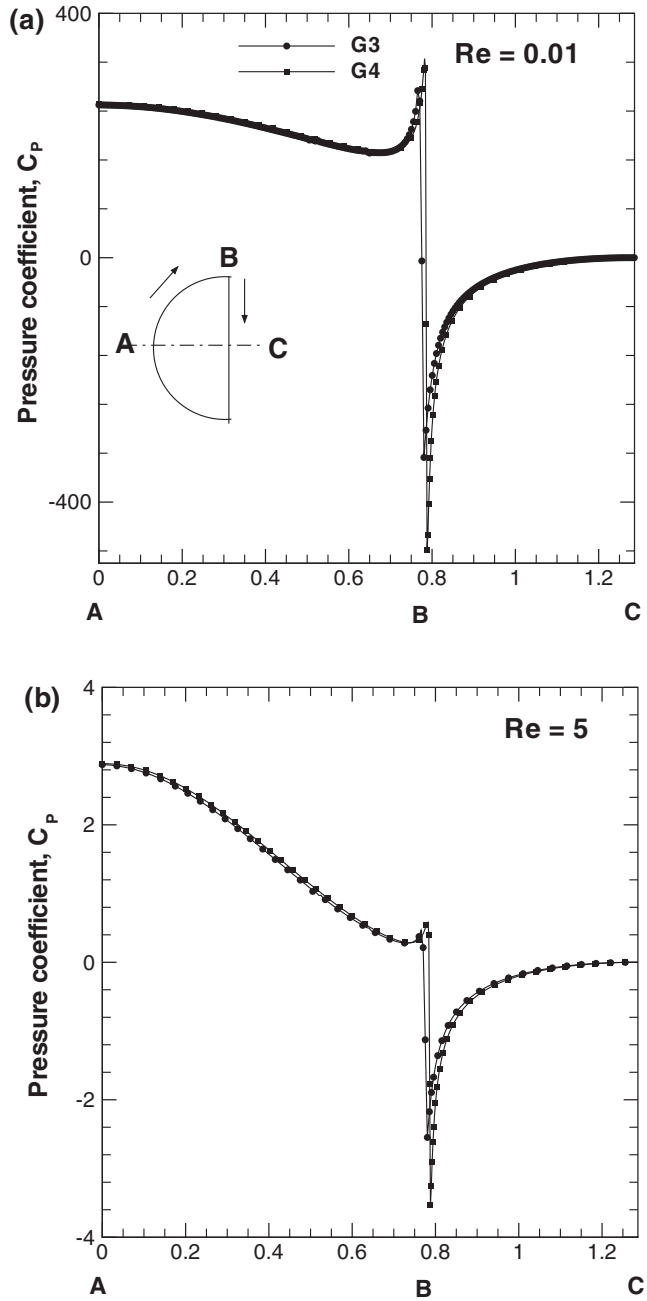

Fig. 3. Variation of pressure coefficients (C_p) with grid $G3$ and $G4$: (a) $Re = 0.01$; (b) $Re = 5$.

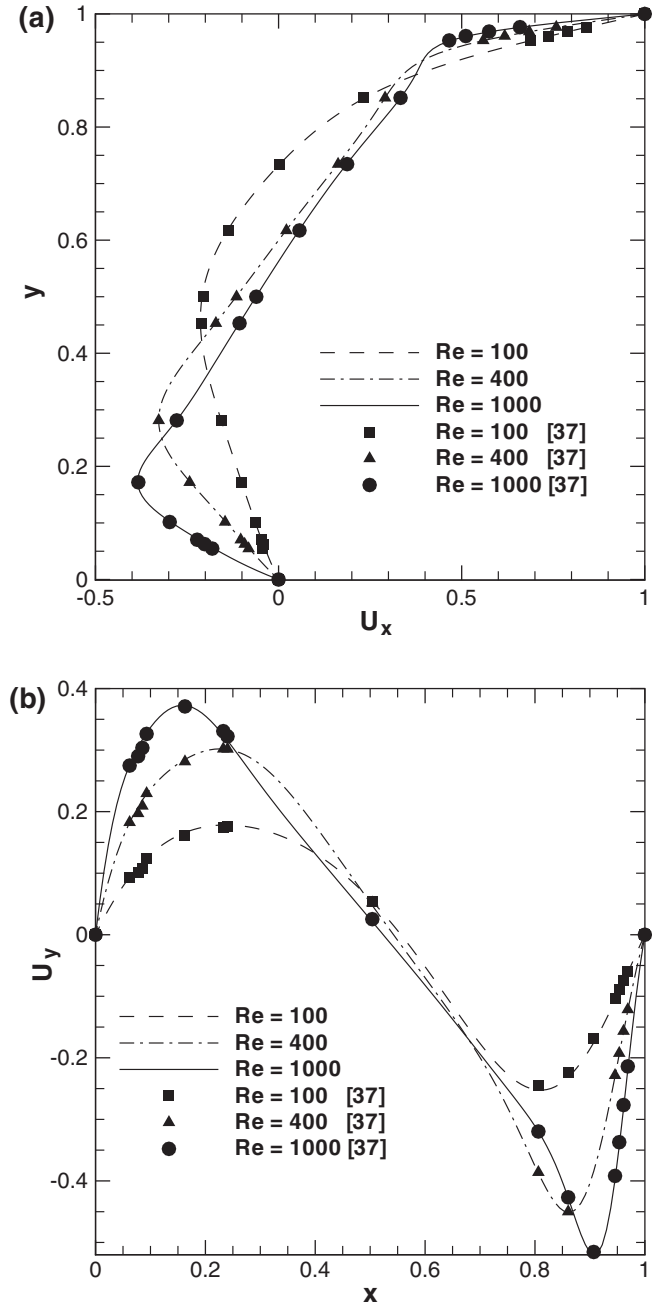

Fig. 4. Comparison of centerline velocity: (a) x-velocity; (b) y-velocity in the lid driven square cavity flow.

Table 3
Influence of time step (Δt) at $Re = 40$.

Non dimensional time step (Δt)	Lift coefficient ($\pm C_L$)	Strouhal number (St)
0.01	0.0143	0.117
0.001	0.0142	0.117

where f is the frequency of vortex shedding and its value has been extracted from the lift coefficient-time evolution behaviour.

3.3. Pressure coefficient

The pressure coefficient is defined as the ratio of the static pressure to the dynamic pressure. On the surface of the cylinder, it is computed using the following expression:

Table 4
Comparison of present results with literature for elliptic cylinders.

Source	e	C_D	
		$Re = 30$	$Re = 40$
Present results	0.2	1.7882	1.6233
Sivakumar et al. [13]		1.7869	1.6211
Present results	0.5	1.7287	1.5570
Sivakumar et al. [13]		1.7284	1.5567

$$C_p = \frac{p_s - p_\infty}{\frac{1}{2}\rho U_\infty^2} \tag{15}$$

where p_s is the local pressure at a point on the surface of the cylinder, and p_∞ is the reference pressure.

Table 5
Comparison of present results (Nusselt number) at *Re* = 40 with that of Bharti et al. [14].

Pr	*e* = 0.2		*e* = 0.5	
	Present	Bharti et al. [14]	Present	Bharti et al. [14]
1	3.985	3.979	3.973	3.973
10	9.313	9.256	9.034	9.030

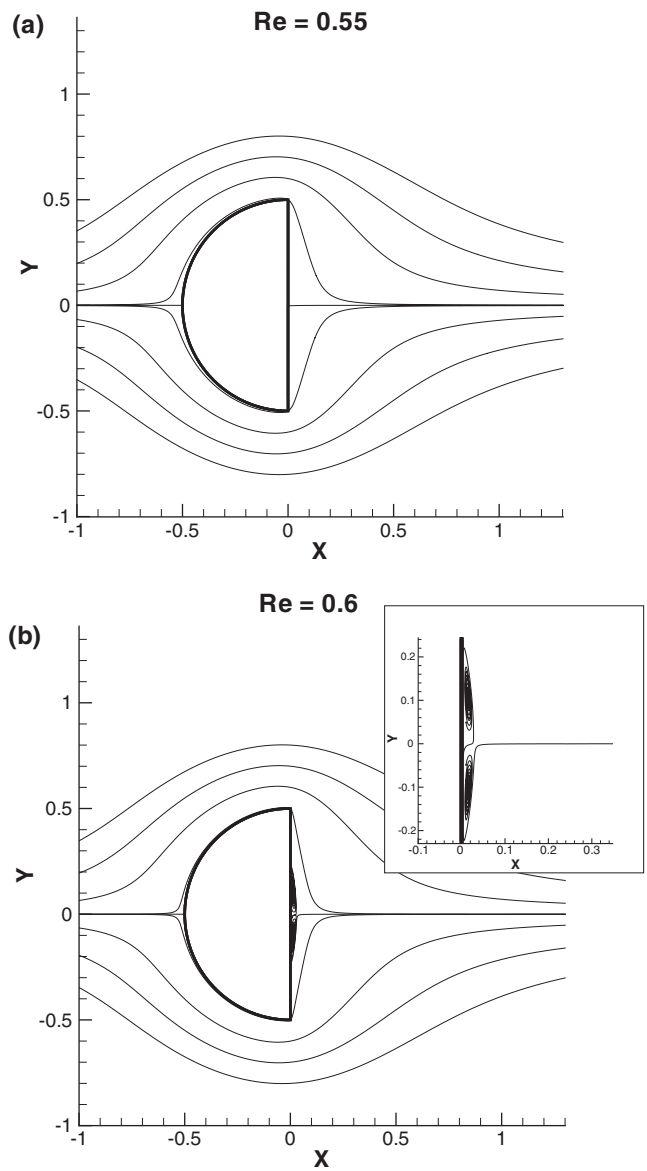

Fig. 5. Streamline profiles: (a) no separation at *Re* = 0.55; (b) flow separation at *Re* = 0.6.

3.4. Vorticity

It is a measure of the local rate of rotation and is defined as the curl of the velocity vector, *i.e.*,

$$\omega = \frac{1}{2}\left(\frac{\partial U_y}{\partial x} - \frac{\partial U_x}{\partial y}\right) \tag{16}$$

3.5. Recirculation (or wake) length

This is an indication of the size of the recirculation region in the flow direction. It is measured from the rear stagnation point to the point of reattachment for the near closed streamline ($U_x = U_y = 0$) on the line of symmetry ($y = 0$) in the downstream section. It is defined as:

$$L_r = \frac{L}{D} \tag{17}$$

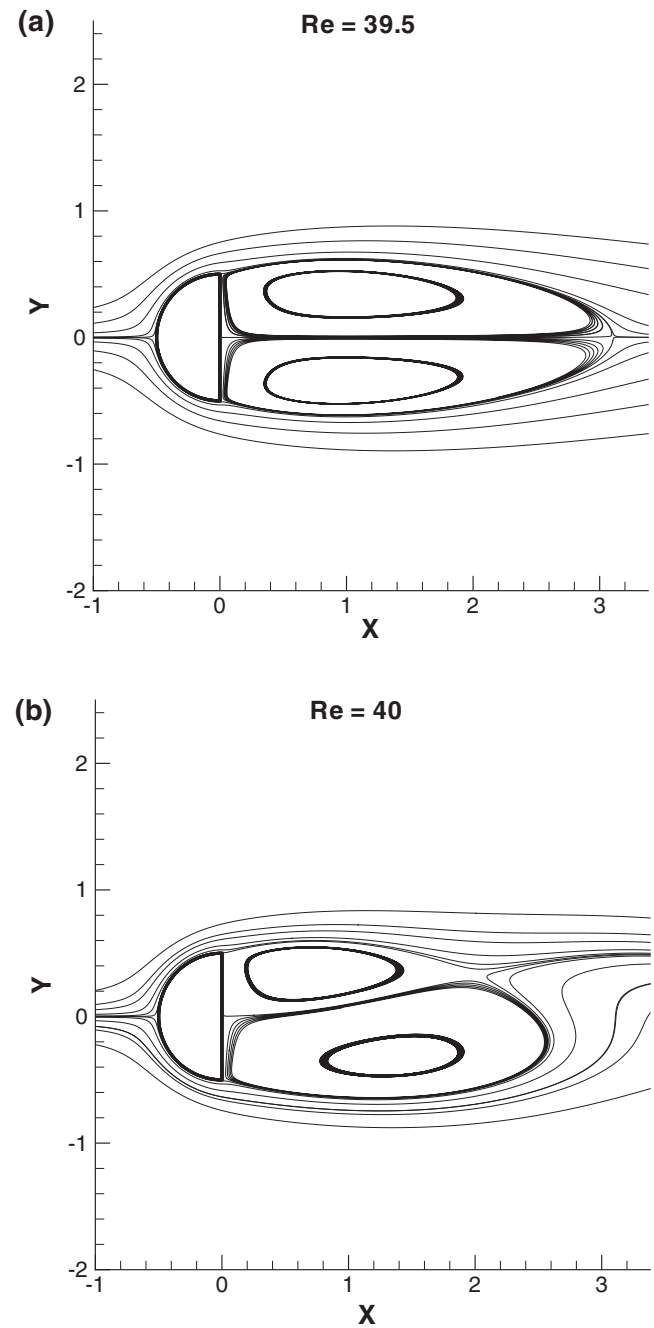

Fig. 6. Streamline profiles: (a) steady flow at *Re* = 39.5; (b) unsteady flow at *Re* = 40.

3.6. Nusselt number

The non-dimensional rate of heat transfer is expressed in terms of the Nusselt number, Nu_L which is defined as:

$$Nu_L = \frac{hD}{k} = -\frac{\partial T}{\partial n_s} \tag{18}$$

While the local Nusselt number provides an indication of how the local rate of heat transfer varies along the surface of the obstacle,

it is the surface averaged value which is frequently needed in process engineering calculations. This is defined as:

$$Nu_{avg} = \frac{h_{avg}D}{k} = \frac{1}{s}\int_s Nu\,ds \tag{19}$$

One of the objectives of the present study is to establish the value of the Reynolds number Re^c, which marks the onset of flow separation. This value was located by examining the value of the stream function and vorticity in the close vicinity of the cylinder. The onset of flow separation corresponds to the change of sign of vorticity as well as it is indicated by some of the velocity vectors being oriented in the opposite direction to that of the mean flow. Similarly, the values of stream function in the fore and aft regions at homologous points were used to ascertain the loss of fore and aft symmetry of the flow field.

The other critical Reynolds number, Re_c, denotes the onset of the vortex shedding regime. This necessitates the solution of time-dependent equations and for this purpose the evolution of lift coefficient with time is monitored. Simulations are continued until either the lift coefficient becomes zero indicating steady flow regime or the flow becomes fully periodic in time. For steady flow, of course, lift coefficient and Strouhal number will be identically zero. Evidently, the flow regime is steady in the range $Re \leqslant Re_c$.

The scaling of the governing equations and the boundary conditions suggests the drag coefficient to be a function of the Reynolds number, whereas the Nusselt number shows further dependence on the Prandtl number. These relationships are explored and developed in this work in the steady flow regime, *i.e.*, $Re \leqslant Re_c$.

4. Numerical methodology

The present numerical investigation has been carried out using FLUENT (version 6.3.26). The unstructured quadrilateral cells of non-uniform grid spacing are generated using the commercial software GAMBIT (version 2.3.16). The grid is chosen to be sufficiently fine to capture the steep gradients in the vicinity of the cylinder. The two-dimensional, steady and unsteady (only for investigation of the critical Reynolds number, Re_c), laminar, segregated solver is used to tackle the incompressible flow and forced convection heat transfer on the collocated grid arrangement. The second order upwind scheme has been used to discretize the convective terms in the momentum and energy equations. The SIMPLE scheme (semi-implicit method for the pressure linked equations) is used for solving the pressure-velocity decoupling. The constant density method was used to input the physical properties of fluid. However, these values are of no particular significance as the final results are presented in a dimensionless form to enhance their potential utility. FLUENT solves the system of algebraic equations using the Gauss–Siedel (G–S) point-by-point iterative method in conjunction with the algebraic multi-grid method (AMG) solver. The use of AMG scheme greatly reduces the number of iterations and CPU

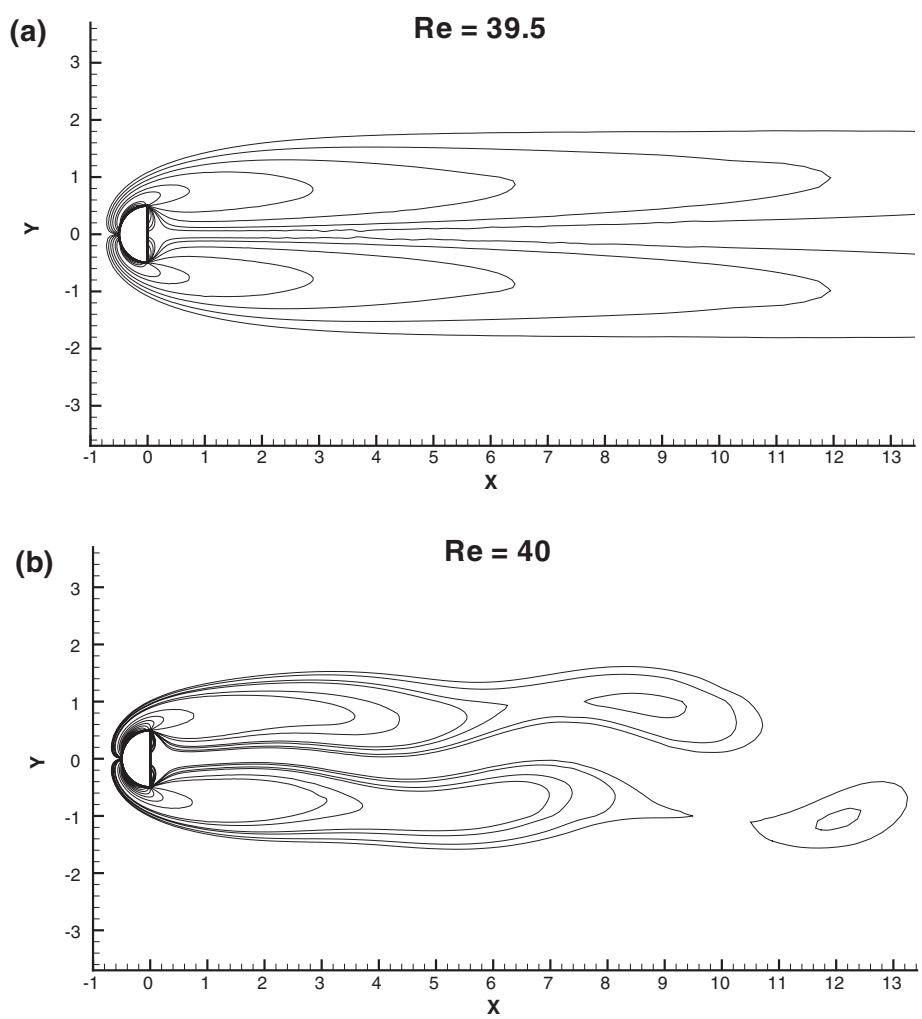

Fig. 7. Vorticity profiles: (a) steady flow at $Re = 39.5$; (b) unsteady flow at $Re = 40$.

time required to obtain the converged solution, particularly when the model equations contain a large number of volumes. The convergence criterion of 10^{-10} is used for the residuals of continuity, x-momentum and y-momentum equations, and of 10^{-15} for the residual of the energy equation. The unsteady solver is used for the identification of Re_c and the criterion of 10^{-8} is used for the residuals of continuity, x-momentum and y-momentum equations. In this case, the non-dimensional time step of 0.01 is used for time-dependent simulations, as detailed in the choice of the numerical parameters section. The unsteady state calculations are stopped, when the lift coefficient reached either a fully periodic steady state or reduced to a sufficiently small value ($\sim 10^{-5}$). Furthermore, a simulation was deemed to have converged when the values of the drag coefficient and Nusselt number had stabilized at least up to four significant digits and/or fully periodic regime has been established.

5. Choice of numerical parameters

Needless to say that the reliability and accuracy of the numerical results is strongly influenced by the choice of numerical parameters, *i.e.*, domain size, grid details, size of time step (Δt) and to some extent on the convergence criterion, etc. It is thus imperative to make prudent choice of these parameters for the final results to be free from numerical artefacts. In this work, the domain is defined by the value of D_∞. Intuitively, it appears that the velocity field decays rather slowly at low Reynolds numbers and therefore it stands to reason that a large value of D_∞ is needed at low Reynolds numbers. The range of the Reynolds number covered in this study is divided into two sub-ranges as $Re < 5$ and $Re \geqslant 5$ and thus, the domain independence study has been carried out separately for these two regions. Four values of (D_∞/D) ranging from 1200 to

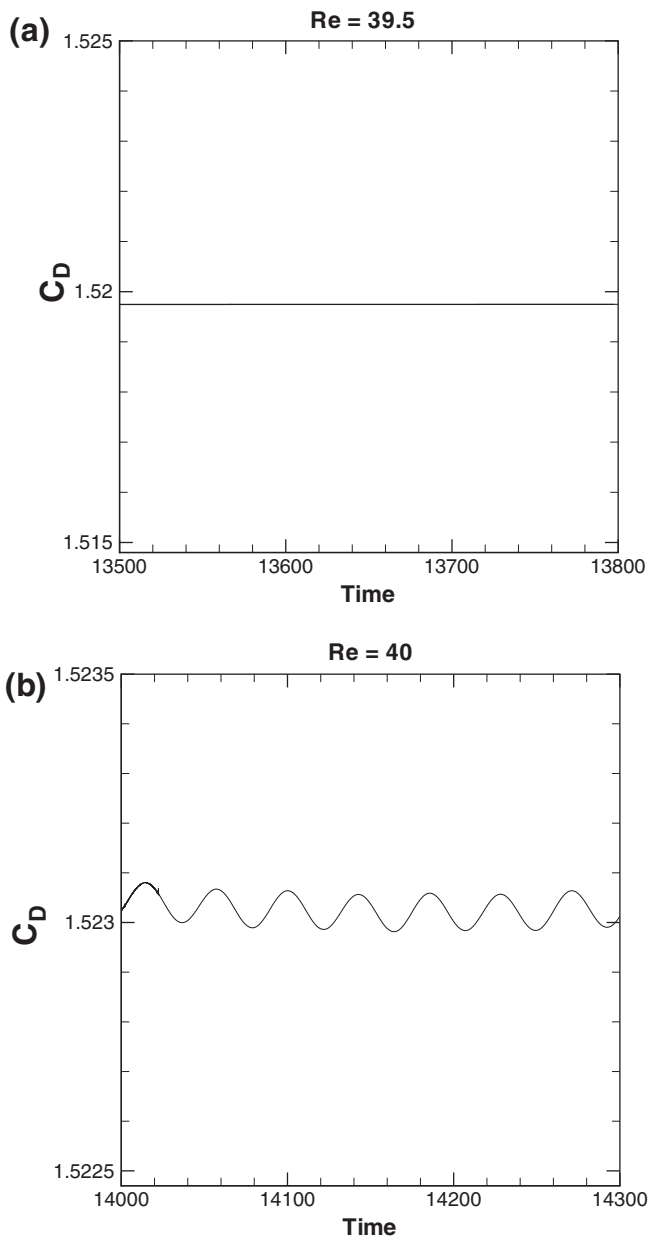

Fig. 8. Evolution of drag coefficient: (a) steady flow at $Re = 39.5$; (b) unsteady flow at $Re = 40$.

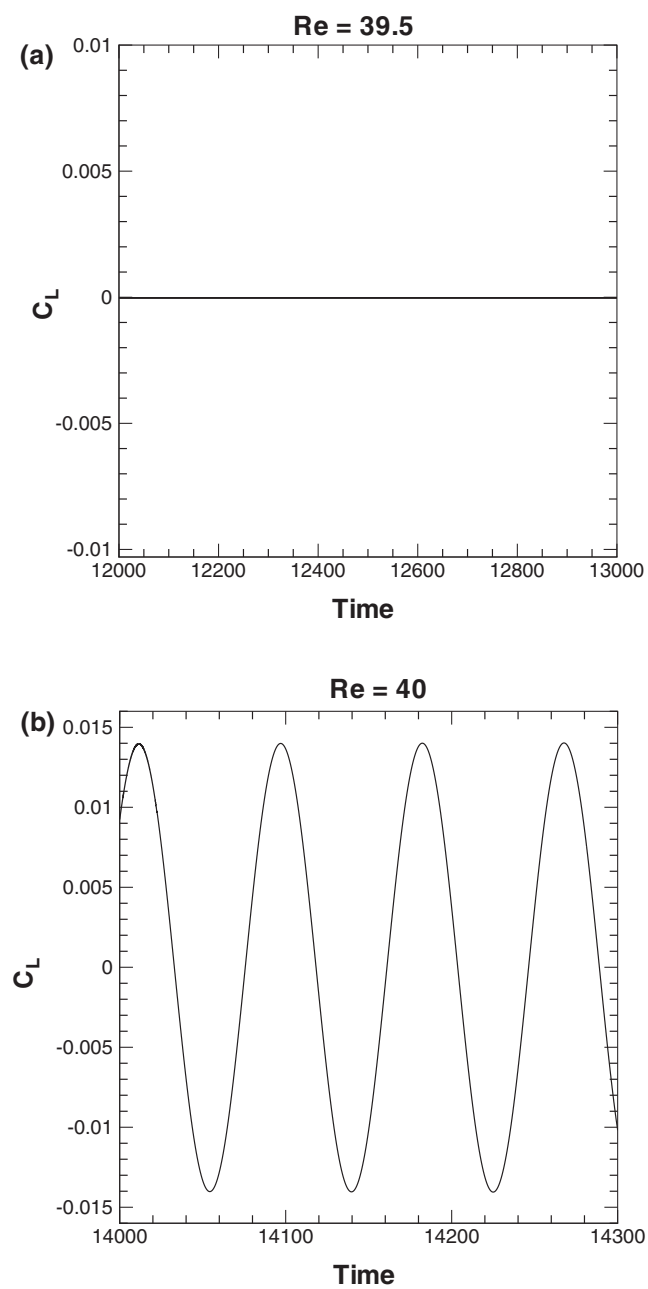

Fig. 9. Evolution of lift coefficient: (a) Steady flow at $Re = 39.5$; (b) unsteady flow at $Re = 40$.

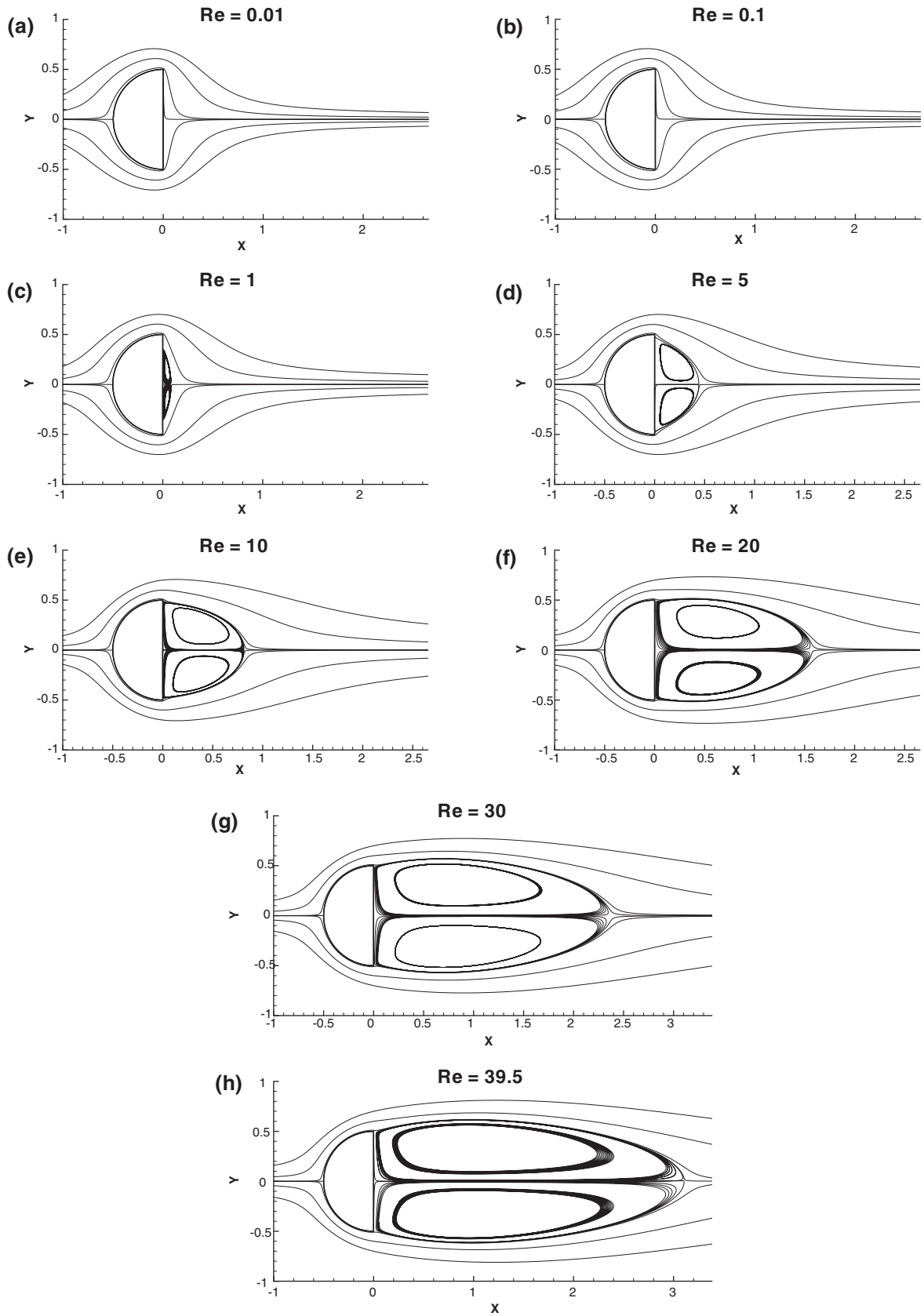

Fig. 10. Streamline profiles at different Reynolds numbers in steady flow regime.

1800 have been used in this work. The domain independence studies have been done with grid $G1$ (detailed in Table 2). The resulting values of the pressure and total drag coefficients are presented in Table 1. An examination of this table shows that at $Re = 0.01$, the values of the individual and total drag coefficients change by less than 1% when the domain size is increased from $(D_\infty/D) = 1600$

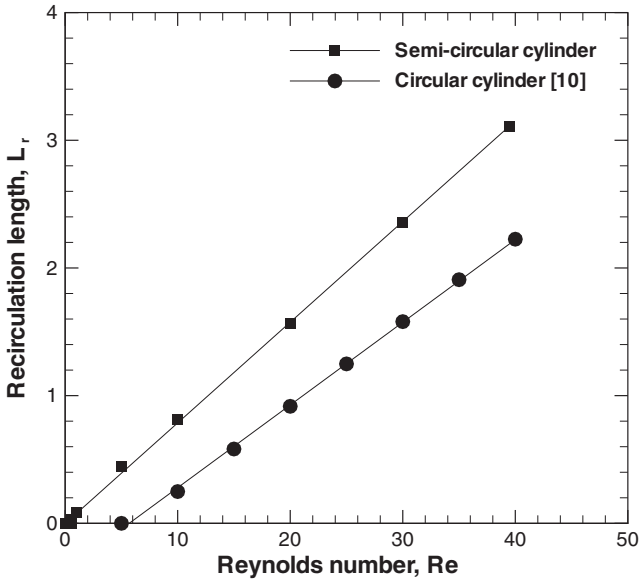

Fig. 11. Dependence of recirculation length (L_r) on Reynolds number.

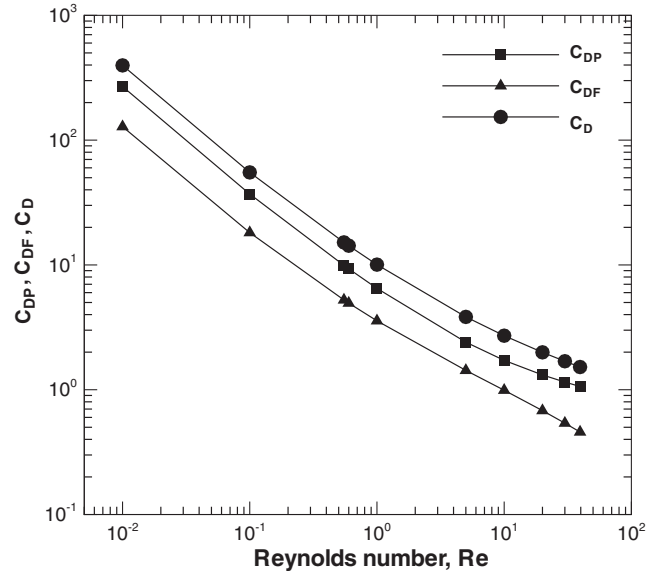

Fig. 13. Variation of drag coefficients (C_{DP}, C_{DF}, and C_D) with Reynolds number.

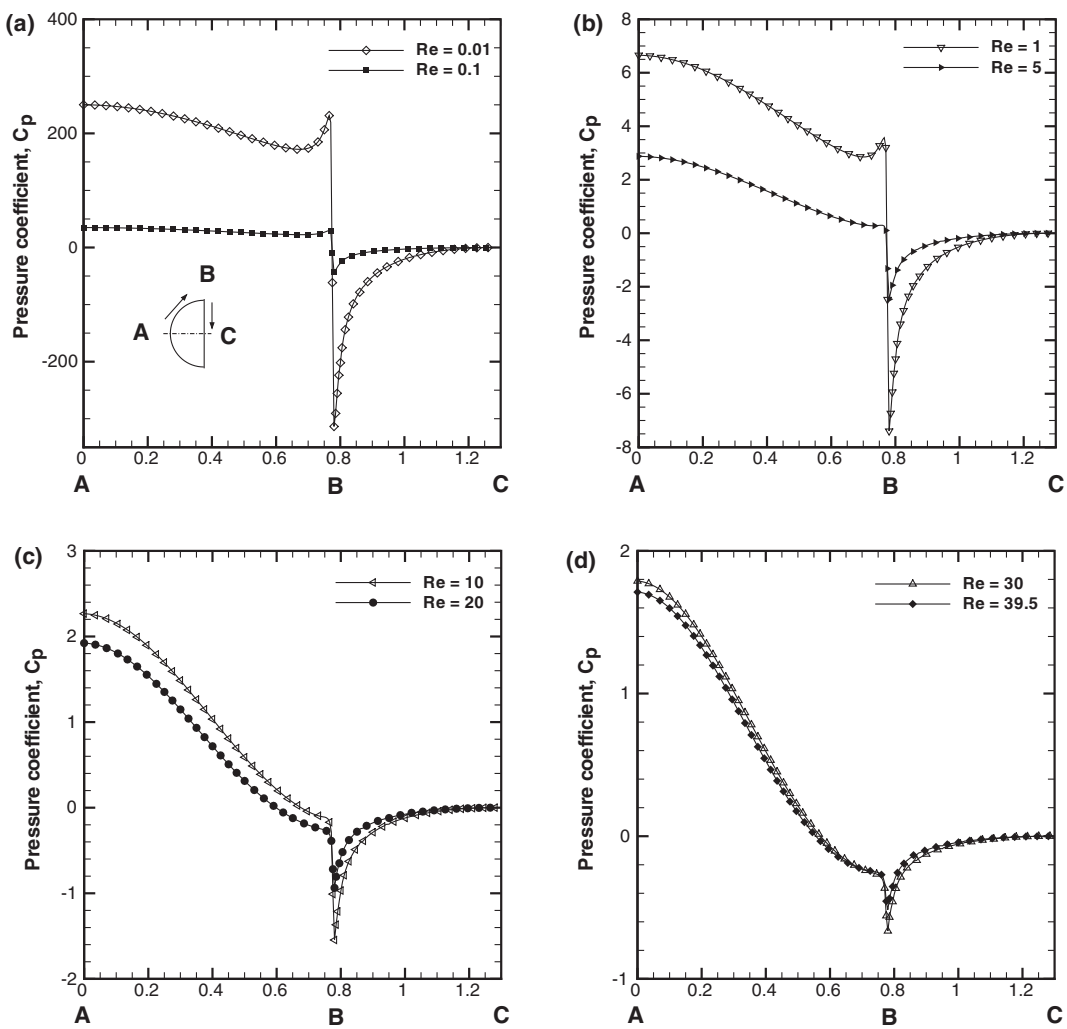

Fig. 12. Variation of pressure coefficient (C_p) with Reynolds number: (a) $Re = 0.01$ and 0.1; (b) $Re = 1$ and 5; (c) $Re = 10$ and 20; (d) $Re = 30$ and 39.5.

to 1800. Undoubtedly, such small changes are accompanied by an enormous increase in CPU time for these two values of (D_∞/D). Therefore, the domain size $(D_\infty/D) = 1600$ has been used in this work for $Re < 5$. Similarly, it appears that $(D_\infty/D) = 1200$ is quite adequate for the range of $Re \geqslant 5$.

Having fixed the domain size, let us now turn our attention to the choice of an optimal grid. An ideal grid should meet two conflicting requirements, namely, it should be sufficiently fine to resolve the steep (velocity and temperature) gradients near the obstacle without, of course, being excessively computational resource intensive. Four grids with varying numbers of control volumes on the surface of the semi-circular cylinder and gradually making it coarser away from the bluff body were created (details are provided in Table 2 and a representative grid is shown in Fig. 2). The resulting values of the individual and total drag coefficients for $Re = 0.01$ and $Re = 5$ are also summarized in this table. An examination of the results at $Re = 0.01$ shows that the value of C_{DP} has stabilized up to about 1.3% and that of C_{DF} within 2.2%, but the total drag results have been stabilized to within 0.2% between grids $G3$ and $G4$. Similarly, examining the results at $Re = 5$, there is very little change in values obtained with $G3$ and $G4$. Furthermore, Fig. 3 shows the variation of pressure coefficient over the surface of the cylinder obtained with grids $G3$ and $G4$. Here also, the two values are seen to be virtually indistinguishable from each other thereby testifying to the adequacy of grid $G3$. Bearing in mind the manyfold increase in CPU time associated with the use of $G4$, it appears that $G3$ denotes a good compromise between the accuracy on one hand and the required computational effort on the other hand. Thus, the results reported herein are based on the use of the grid $G3$.

Since the identification of the critical Reynolds number (Re_c) marking the onset of the laminar vortex shedding regime necessitates the solution of the time-dependant equations, an appropriate choice of the time step, Δt, is also required. Table 3 shows the values of the lift coefficient and Strouhal number obtained with two values of $\Delta t = 0.01$ and $\Delta t = 0.001$ at $Re = 40$, the maximum value of the Reynolds number used in this study. The two values are seen to be virtually indistinguishable from each other, and therefore, the (non-dimensional) value of $\Delta t = 0.01$ has been used in this work. A second order scheme has been used to march in time.

In summary, the numerical results reported in this work are based on the following choice of the numerical parameters: grid $G3$; $\Delta t = 0.01$; $D_\infty/D = 1600$ for $Re < 5$ and $D_\infty/D = 1200$ for $Re \geqslant 5$.

6. Results and discussion

Before undertaking the detailed presentation of the new results for a semi-circular cylinder, it is deemed important to establish the general validity of the numerical solution approach used here. This, in turn, will facilitate ascertain the accuracy and reliability of the new results obtained herein. For this purpose, two bench mark problems have been solved. The first one concerns the laminar flow in a 2-D lid driven square cavity for which very accurate results are available in the literature [37]. A representative comparison between the x-velocity and y-velocity at the centerline are compared with that of Ghia et al. [37] in Fig. 4, for three values of the Reynolds number. The two values are seen to be in excellent agreement; the maximum difference being of the order of ~2%. The second bench mark problem studied here is the flow over an elliptical cylinder of aspect ratio, $e = 0.2$ and $e = 0.5$. Tables 4 and 5 show comparisons between the present results and that of Sivakumar et al. [13] for drag and of Bharti et al. [14] for Nusselt number. Once again, the correspondence is seen to be excellent. It is also worthwhile to add here that more detailed and extensive comparisons with the other literature studies are available in Sivakumar et al. [13] and Bharti et al. [14]. Based on these comparisons together with our past experience, the new results for the semi-circular cylinder reported herein are believed to be reliable to within 1.5–2%.

6.1. Structure of flow field

6.1.1. Fluid flow phenomena

It is customary to visualize the flow in terms of streamline contours. As stated earlier, one of the objectives of the present work is to determine the value of the Reynolds number corresponding to the onset of flow separation (Re^c) and the onset of laminar vortex shedding regime (Re_c). The value of Re^c was ascertained by examining the streamline patterns and vorticity values close to the surface of the cylinder. At low Reynolds numbers ($Re \to 0$), fluid elements are able to follow the surface of bluff body. At a critical Reynolds number, an adverse pressure gradient occurs at some point in the rear of the cylinder and the flow detaches itself from the surface of the cylinder. At this point, vorticity also changes its sign and in fact, some of the velocity vectors also point in the direction opposite to the direction of mean flow. All these tools have been used in this work to ascertain the value of Re^c. Fig. 5 shows the streamline profiles for $Re = 0.55$ and $Re = 0.6$. At $Re = 0.55$, the flow appears to be attached to the surface of the cylinder whereas at $Re = 0.6$, a bubble is clearly visible. Therefore, the value of Re^c lies somewhere in the range 0.55–0.6. No further resolution was attempted to pin point precisely this value. Since the two values are only about 8% different from each other, the mean of the two is designated as the value of Re^c in this case. This value is almost an order of magnitude lower than that for a circular cylinder ($Re^c \sim 4$–5). On the other hand, the value of Re^c of 0.58 is also lower than that for a square cylinder for which a value of $Re^c \sim 1.2$ has been reported recently [9]. While the lower value for a square cylinder than that for a circular cylinder is attributed to the reduced degree of streamlining, what needs to be recognised here that both circular and square cylinders exhibit symmetry about the y-axis, the semi-circular cylinder does not. Therefore, it is plausible that the reduced degree of streamlining together with the above-noted loss of symmetry contribute to the early flow separation which is also reflected in poor pressure recovery on the flat surface, even at $Re = 0.01$ (Fig. 3(a)).

Similarly, Figs. 6–8 show the gradual development of the transition from the flow being steady to unsteady. At $Re = 39.5$, the streamline patterns confirm the symmetry and stable nature of the separated flow region about the $y = 0$ plane which is clearly destroyed at $Re = 40$. The vorticity contours plotted in Fig. 7 also show that no shedding occurs at $Re = 39.5$ while at $Re = 40$, one can clearly see the commencement of vortex shedding. As the Reynolds number is gradually incremented, the inertial forces tend to destabilize the flow. In experimental studies, vibration of equipment, fluctuating voltage, imperfect boundaries, etc. can induce perturbation in the flow. Obviously, none of these factors are relevant

Table 6
Present numerical values of drag coefficient.

Re	C_{DP}	C_{DF}	C_D
0.01	268.94	128.190	397.13
0.1	36.99	18.147	55.133
0.55	9.918	5.238	15.155
0.6	9.310	4.943	14.253
1	6.489	3.566	10.055
5	2.404	1.429	3.832
10	1.719	0.991	2.710
20	1.313	0.680	1.993
30	1.152	0.540	1.691
39.5	1.062	0.458	1.520

in numerical studies. On the other hand, numerical diffusion, truncation errors, etc. can reach significant magnitudes to destabilize the flow. Evidently, at low Reynolds numbers, the dominant viscous forces prevent these perturbations from growing and therefore these subside with time and the flow retains its steady character. On the other hand, as the Reynolds number is gradually incremented, viscous forces are weakened and the increasing inertial forces can accelerate the loss of steadiness. This is qualitatively borne out by the results shown in Figs. 6–8 indicating that how a small increase in the value of the Reynolds number can tilt this balance between the viscous (stabilizing) and inertial (destabilizing) forces present in the flow field. Finally, Figs. 8 and 9 show the evo-

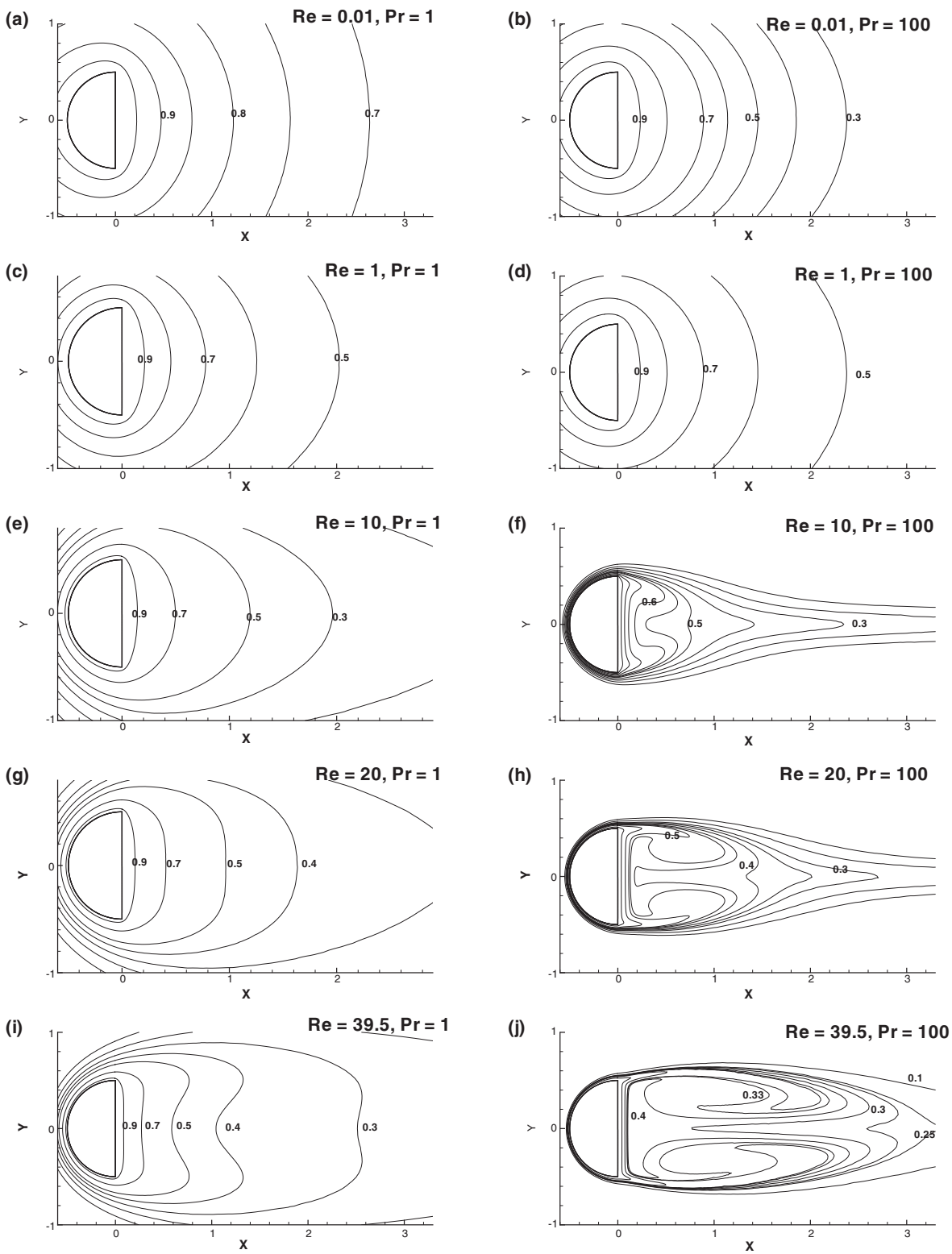

Fig. 14. Isotherm profiles at various Reynolds numbers for $Pr = 1$ and $Pr = 100$.

lution of drag coefficient and lift coefficient with time which is seen to be steady at $Re = 39.5$, while it exhibits fully periodic behaviour in time at $Re = 40$. As expected, the time-period of drag coefficient is half of that of the lift coefficient. The corresponding values of the characteristic parameters are: $St = 0.117$, $\overline{C}_D = 1.523$, $C_L = \pm 0.014$ at this point, *i.e.*, at $Re = 40$. In nut shell, these trends clearly suggest that the critical Reynolds number, Re_c, denoting the cessation of the steady flow regime to lie in the range 39.5–40. As expected, this value is slightly lower than the oft-cited value of 46–47 for a circular cylinder and $Re_c = 45$ for a square cylinder [9] for the reasons advanced earlier in the context of Re^c. The present value of $Re_c = 39.5$–40 is also consistent with the observation of Sophy et al. [31] who reported the flow to be time-dependent at $Re = 65$.

Fig. 10 shows streamline profiles for a range of Reynolds numbers encompassing the region $Re \leqslant Re_c$ to visualize the gradual changes in the flow patterns. Fig. 10(a) and (b) represents the flow at extremely low values of the Reynolds number without any separation, *i.e.*, streamlines are able to follow the contour of the body, and the pressure gradient everywhere on the surface is favorable. On the other hand, as the Reynolds number is slightly increased, adverse pressure gradient is established on the rear plane surface leading to flow separation. Fig. 10(c)–(h) shows the separated flow regions in the range of $Re^c \leqslant Re \leqslant Re_c$. The separation bubble grows with the Reynolds number almost linearly as seen in Fig. 11. At the same Reynolds number, the bubble size (wake region) for the semi-circular cylinder is larger than that for a circular cylinder [10]. The linear relationship between the Reynolds number and recirculation length can be best represented by the following expression for the Reynolds number range of $1 \leqslant Re \leqslant 39.5$ with the average error of 2.2%, which rises to the maximum of 6.4%

$$L_r = 0.079Re \tag{20}$$

Some further insights into the nature of flow can be gained by examining the pressure coefficient variation on the surface of the cylinder. Since the flow is symmetric about mid-plane, Fig. 12 shows the dimensionless pressure distribution over the half of the cylinder (traversing in the clockwise direction). The pressure distribution on the curved surface is seen to be qualitatively similar to that seen on the front surface for a circular cylinder, albeit at point B, the pressure is negative for the cases when the flow has separated. At low Reynolds numbers ($Re < Re^c$), the viscous forces dominate the flow, surface pressure varies very little along the surface between points A and B, it suddenly drops at B, followed by a partial recovery in the rear (between points B and C). The pressure coefficient at the front stagnation point decreases with the increasing Reynolds number.

6.1.2. Drag coefficient

Fig. 13 shows the variation of the individual and total drag coefficients with the Reynolds number while Table 6 summarizes their numerical values. The results shown here adhere to the classical

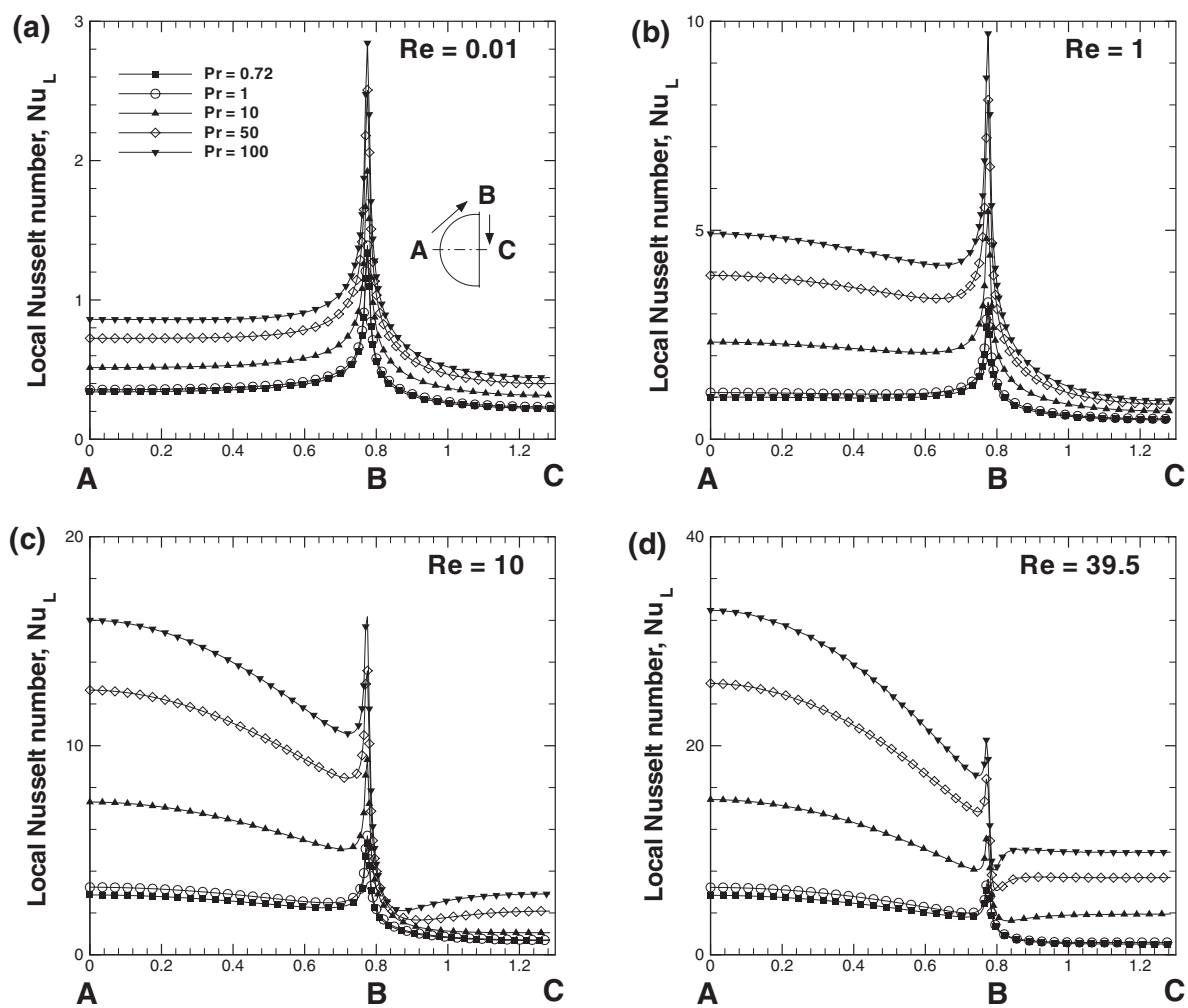

Fig. 15. Local Nusselt number (Nu_L) at (a) $Re = 0.01$; (b) $Re = 1$; (c) $Re = 10$; (d) $Re = 39.5$.

inverse dependence of drag coefficient on the Reynolds number. Irrespective of the value of the Reynolds number, the total drag coefficient draws a significant contribution from the pressure drag coefficient (~70%), and the extent of this contribution varies very little with the value of the Reynolds number. For instance, at $Re = 0.01$, the ratio (C_{DP}/C_D) is 0.68 whereas at $Re = 39.5$, it is (C_{DP}/C_D) = 0.7, thereby suggesting that the viscous drag is much smaller than the form drag over the range of conditions studied in this work.

6.2. Forced convection heat transfer

The forced convection heat transfer characteristics have been explored for the steady symmetric flow regime ($0.01 \leqslant Re \leqslant 39.5$) and for scores of values of the Prandtl number as $Pr = 0.72$, 1, 10, 50, and 100. The constant wall temperature (CWT) condition has been applied at the surface of the cylinder. It is useful to visualize the thinning of the thermal boundary layer with the increasing values of the Prandtl and/or Reynolds numbers. Furthermore, isotherm contours also help identify the regions of locally "high" or "low" temperature, which may be an important factor in the processing of temperature sensitive materials. Fig. 14 shows the representative isotherm profiles in the vicinity of the cylinder for a range of combinations of the values of the Reynolds and Prandtl numbers. The clustering of isotherms on the front surface of the cylinder with the gradually increasing value of the Prandtl number at a constant value of the Reynolds number indicates the progressive thinning of the thermal boundary layer, which also suggests the Nusselt number to be maximum at the front stagnation point. At low Reynolds numbers, convection is weak and the isotherms contours are seen to be almost circular and extend farther away which gradually are distorted at high Reynolds numbers due to the increasing advection and due to flow separation. At $Re = 1$ and $1 \leqslant Pr \leqslant 100$, the recirculation zone is small and as such has negligible effect on the shape of the isotherms. The irregular pattern of isotherms at $Re = 10$ for $Pr = 100$ is due to the enhanced flow recirculation in the vicinity of the cylinder and it forms "wake-type" structures at $Re = 20$ and 30.

By writing heat balance at a point on the surface of the cylinder, it is easily seen that the local value of the Nusselt number is governed by the local temperature gradient and in view of the maximum temperature gradient at the front stagnation point, one would expect the Nusselt number to be maximum at this point and it will gradually decrease along the surface. The variation of the local Nusselt number (Nu_L) along the surface (A–B–C) is shown in Fig. 15 for four values of the Reynolds number and scores of values of the Prandtl number. As noted above, the Nusselt number is seen to be highest here and it progressively decreases along the surface until the point B is reached. Due to the singularity here, the Nusselt number goes through another maximum value, before falling rapidly in the rear of the cylinder. While the value of the Nusselt number both at the front stagnation point A and at corner point B increases with both the Reynolds and Prandtl numbers, the location of the maxima of local Nusselt number (for $Pr \geqslant 10$) is shifted from point B to point A as the Reynolds number is gradually increased. On the other hand, there is a very little variation in the value of the Nusselt number along the part BC of the rear flat surface.

In process engineering calculations, reliable values, of average heat transfer coefficient are frequently needed. This is easily evaluated by averaging the values such as that shown in Fig. 15 over the surface of the cylinder, Eq. (19). In order to ascertain the importance of the relative contribution of the curved front and flat rear surfaces, the corresponding average values are plotted in Fig. 16. Naturally, while not only the local values in the front and rear parts of the semi-circular cylinder are different, but the areas for heat transfer also differ by a factor of π, the bulk of the total heat transfer occurs from the curved surface. Broadly, the average heat transfer coefficient increases with the both Reynolds and Prandtl numbers. Needless to say here that since $Pr > 1$ (except for one value), the thermal boundary layer will always be thinner than the

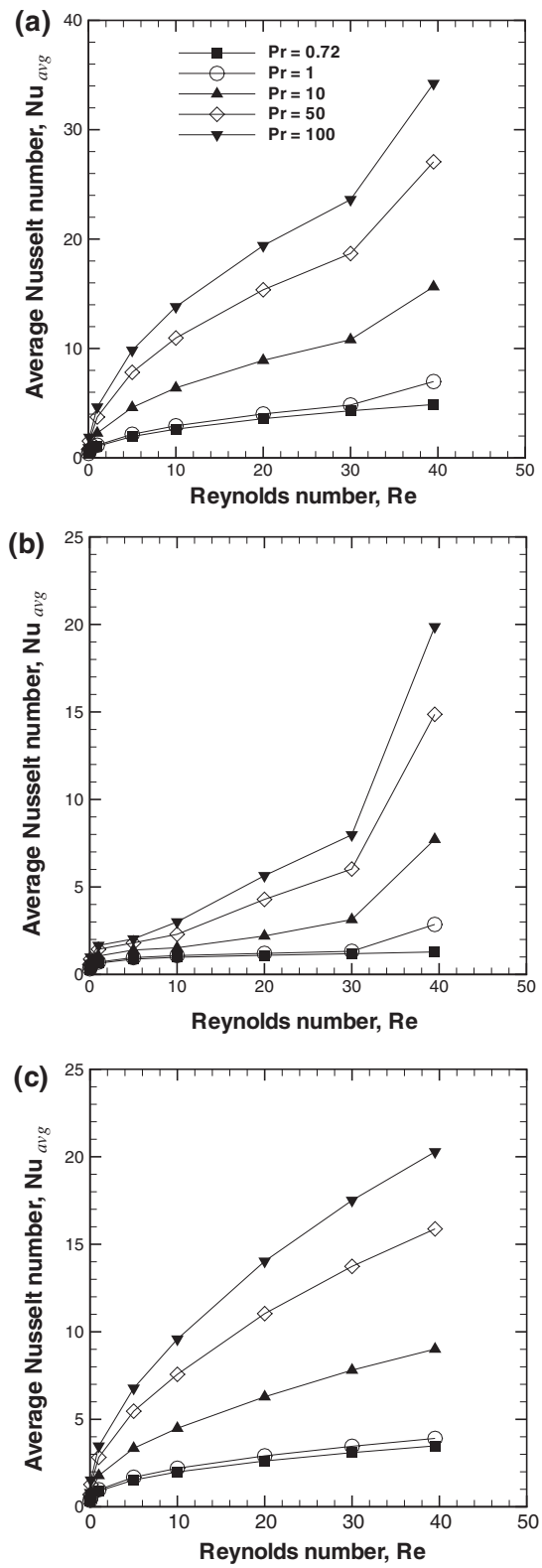

Fig. 16. Variation of average Nusselt number (Nu_{avg}) with Reynolds number (Re) and Prandtl number (Pr): (a) for curved surface; (b) for flat surface; (c) for full cylinder.

corresponding momentum boundary layer. Furthermore, the thermal boundary layer becomes thinner and thinner with the increasing values of the Reynolds and/or Prandtl numbers. This is tantamount to the lowering of the resistance (to heat transfer $\sim\delta_T/k$) thereby resulting in a positive dependence of the Nusselt number on both *Re* and *Pr*. Indeed, this assertion is borne out by the results shown in Figs. 15 and 16. Furthermore, the boundary layer analysis for a flat plate suggests the rate of change of δ_T with the Reynolds number ($\sim Re^{1/2}$) to *Re* be faster than that with the Prandtl number ($\sim Pr^{1/3}$), it is reasonable to postulate that the Nusselt number shows a stronger dependence on the Reynolds number than that on the Prandtl number. For instance, at $Pr = 100$, the value of Nusselt number increases from $Nu = 3.47$ to $Nu = 20.3$ as the value of the Reynolds number is increased from $Re = 1$ to $Re = 39.5$ which is approximately in accordance with $Nu \propto \sqrt{Re}$. However, it is fair to expect that the scaling of $Nu \sim Re^{1/2}Pr^{1/3}$ is applicable only when modest advection is present. In the present case, this seems to be true for $(Re \cdot Pr) > \sim 5$–10. Conversely, since the chief mode of heat transfer is conduction at small Prandtl numbers (δ_T is large) and/or low Reynolds numbers, the dependence of the Nusselt number is expected to be weaker than $\sqrt{Re}$. In the range $0.01 \leqslant Re \leqslant 1$, it is found approximately to be $Nu \sim Re^{1/4}$. Furthermore, the present heat transfer results shown here suggest the scaling of $Nu \sim Pr^{1/3}$, especially for $Re \geqslant 5$.

7. Conclusions

In this work, the flow over and heat transfer from a semi-circular cylinder immersed in Newtonian fluids has been studied numerically. The critical Reynolds numbers denoting the onset of flow separation and the onset of the laminar vortex shedding regime have been identified. The first transition, *i.e.*, the onset of flow separation occurs somewhere in the range $0.55 < Re^c \leqslant 0.60$. The second transition corresponding to the cessation of the steady flow regime is seen to occur somewhere in the range $39.5 < Re \leqslant 40$. Since the interval of these values is rather small, one can use the mean values of $Re^c = \sim 0.575$ and $Re_c = \sim 39.75$ as their critical values. For the latter transition, the corresponding values of the Strouhal number (*St*) and time-averaged drag coefficient are found to be 0.117 and 1.523 respectively at $Re \simeq 40$. Owing to the flat rear surface, both these transitions occur at lower Reynolds numbers than that for a circular cylinder and a square cylinder, which is due to the reduced streamlining and lack of symmetry about the *y*-axis of the bluff body. Furthermore, extensive results on flow phenomena and heat transfer have been obtained in the steady flow regime, i.e., $Re < Re_c$. Useful physical insights into the nature of flow and temperature fields are provided in terms of streamlines, surface pressure and isotherm contours. The size of the wake region grows almost linearly with the Reynolds number. The total drag is dominated by the pressure contribution even at low Reynolds numbers. Similarly, the heat transfer results conform to the expected positive dependence on the Reynolds and Prandtl numbers. The bulk of the heat transfer occurs from the front curved surface of the cylinder. This study is restricted to the two-dimensional flow regime and also the thermo-physical properties have been assumed to be constant thereby limiting the applicability of the present results to the situations where $\Delta T = (T_w - T_\infty)$ is small and one can thus assign mean values to the physical properties. Future studies will hopefully address some of these as well as many other issues related to this relatively unexplored flow configuration.

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